Slambient Master Blueprint: The 4-Layer Hybrid Ambient Workstation (Complete Production Edition)

This master architecture document locks in all finalized design elements, polyphony budgets, generative logic, synesthetic color-mapping UI themes, UI layout specifications, production-level reliability features (state persistence, brickwall limiting, MIDI clock sync), preset management, and plug-and-play instant generative modes for the custom **Slambient** (Sloop-FM1) firmware project.

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1. Aesthetic, Sonic Goals & Maestro Inspirations

- **The Vibe:** Warm, nostalgic, cinematic "Boards of Canada" ambient aesthetic with organic drift, emotional depth, and zero digital popping/clipping.
- **Analog Imperfections:** Built-in tape flutter / wow-and-flutter modulation running across the master output chain.
- **Dynamic Breathing & Dynamic Ducking:** Designed with natural pauses, velvet-smooth attacks, and soft rolling envelopes so the "wall of sound" blooms and rests organically. Subtle dynamic ducking (-3 dB to -6 dB) is automatically applied to the sub-drone and shimmer arpeggio during active lead phrases, maximizing harmonic clarity and space without adding extra voice burden.
- **Brian Eno Engine (Generative Phase-Locked Ecosystems & Coprime Architecture):**
    - **Base Master Clock Unit & Exact Layer Dividers:** The master clock tick unit is explicitly defined as a 16th note (250 ms at 60 BPM). Exact layer dividers operate off this base clock: Layer 1 (Euclidean percussion) runs at a 1-tick step rate with a 7-step loop period (7 ticks = 1.75s); Layer 3 (Shimmer Arp) runs at a 1-tick step rate with an 11-step loop period (11 ticks = 2.75s); Layer 2 (Chords/Lead) runs at a 2-tick step rate (8th notes) with a 13-step loop period (26 ticks = 6.5s); Layer 4 (Sub-Drone) runs at a 16-tick step rate (whole notes/bars) with a 17-step loop period (272 ticks = 68s). Macro cycles utilize a 19-step rhythm rotation and a 23-step Weather mutation trigger. These extended coprime clock phase periods represent clock phase cycle durations rather than a mathematical guarantee against melodic motif repetition.
    - **High-Entropy Phase Evolution, Weighted Random Walk & "Weather" Macro Layer:** High-entropy phase evolution with low repetition probability across multi-hour sessions is achieved by combining coprime multi-layer step cycles, layer clock dividers, and a weighted random walk with tonic/dominant gravitational pull (where tension events—defined as melodic note intervals outside the pentatonic consonance set of the active mode, such as tritones or minor 2nds—bound discordance to a 10% statistical long-run target across multiple seeds and modes). A 20–40 minute ultra-slow drift "Weather" macro

layer modulates global density and harmonic behavior over continuous multi-hour sessions.

- **Hiroshi Yoshimura Engine (Kankyo Ongaku Fluidity):**
    - **Environmental Micro-Glides (Thermal Portamento):** Fluid, variable-speed portamento curves that make harmonic shifts bend and "melt" naturally like water currents rather than stepping abruptly.
    - **Acoustic Spatial Diffusion:** Algorithmic feedback and stereo expansion scaling dynamically with drone spectral weight to simulate physical architectural space.
- **Four Tet Textures:** Acoustic-to-pad hybrid transient envelopes pairing woody plucks with lush washes.
- **Aphex Tuning & Micro-Timing:** Alternative micro-tunings, sub-millisecond event offsets within 64-sample audio blocks (~1.45 ms at 44.1 kHz) for sub-millisecond micro-timing humanization, and micro-timing swing for organic human feel.

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2. The 4-Layer Slambient Architecture

To maximize the FM-1 hardware without cluttering the frequency spectrum, the system is architected into three decoupled subsystems running off a unified master clock, while the sound engine is divided into four distinct, concurrent layers:

- **Subsystem 1: Generative Engine:** Manages 6-prime clock dividers (7, 11, 13, 17, 19, 23 steps operating off the 16th-note master tick), scale math, weighted random walk, mood presets, and seeded event scheduling.
- **Subsystem 2: Existing Synth & Sequencer Engine:** Handles track routing, note event execution, voice allocation, amplitude envelopes, and live performance interaction on the 27 silicone keys (where white keys 1–16 function as musical keys with OCT-/OCT+ transposition, while physical navigation buttons SEQ, EDIT, FX, GLO, and HOME retain menu navigation).
- **Subsystem 3: Optional Audio DSP & UI:** Manages master bus tape effects (wow/flutter), provisional reverb, lookahead peak limiting, 24 FPS strip-rendered display output, preset management, and two-sector flash journaling persistence.

1. **Layer 1: Euclidean Rhythms (The Pulse)**
    - Generates organic, syncopated rhythmic percussion and micro-accents.
    - Uses ultra-short percussive envelopes that decay instantly so they never steal voices from the pads.
2. **Layer 2: Ambient Chords & Lead Engine (The Harmony & Voice)**
    - **Weighted Random Walk & Tonic Gravity:** Discards rigid loops in favor of a weighted random walk driven by tonic/dominant gravitational pull and interval probability distributions across scale degrees (Dorian/Lydian modal shifts), anchoring pitch selection to tonal centers.

    - **Call-and-Response Interplay:** Meandering lead phrases trigger *first* with clear presence, followed by a breathing pause and a responsive pad swell anchored to that musical space.
    - **Controlled Discordance & 10% Tension Rule Target:** The 10% tension rule is defined as a long-run statistical target with tension defined as non-diatonic or passing scale degrees outside the primary consonance set. Tightly bounded interval step probabilities ensure that tension resolves harmonically back to root stability without artificial thresholding.
3. **Layer 3: Micro-Arpeggio Generator (The Shimmer)**
    - An independent, fast-moving melodic layer that trickles notes in a randomized, non-repeating high-mid cascade.
    - Shaped with delicate velocities to sound like twinkling mist or falling water over the pad floor.
4. **Layer 4: Living Sub-Drone Bed (The Tectonic Floor)**
    - A continuous, slow-evolving spectral bed that morphs its harmonic balance over minutes rather than seconds.
    - Tracks the active root anchor and slowly swells sub-bass overtones to keep emotional tension grounded.

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3. Plug-and-Play Instant Ambient Mode ("Instant Slambient")

- **Zero-Thought Operation:** A dedicated startup mode allowing you to simply turn on the FM-1, press play, and let the device fully build out a complete, self-sustaining ambient world instantly.
- **Mood Presets:** Pre-baked macro profiles (e.g., *Nostalgic Drift, Glacial Morning, Hauntological Tape*) that automatically configure scale parameters, drone weight, Euclidean density, and arpeggio speed depending on the desired mood.

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4. Polyphony Management, Audio Safety & Memory Constraints

- **Generative Co-Processor Architecture:** Slambient operates as an internal Generative Event/MIDI Co-Processor triggering notes, CCs, and envelopes into the stock SLOOP/Felucca synthesis and drum engine (Tracks 1–4) rather than attempting to synthesize PCM waveforms from scratch inside the high-priority audio ISR. The pre-computed `SINE_LUT[1024]` lookup table is used solely for low-frequency control signals (LFOs), tape wow/flutter delay modulation, thermal portamento pitch curves, and synesthetic UI polygon rendering.
- **DSP Asymmetry Principle (Drum vs. Synth Voice Cost):** Drum voices and synth voices run on completely separate DSP pipelines and are not 1:1 interchangeable in CPU cycles. Drum voices are lightweight one-shot triggers with short decay (50–200

ms), whereas synth voices (especially 6-op FM6 and 4-pole resonant VA) require intensive per-sample DSP calculations.

- **Authoritative Voice Allocation Table & Reserved Ownership (8 Synth Voices + 4 Drum Voices):** Reserved voice ownership is structured across an authoritative allocation table. The cross-track 8-voice allocator is designated as a design hypothesis to be verified in `voice.c` during the Step 2 source audit, with strict per-track partitioning reserved as the fallback:
    - **Voice 0 (Track 1):** Sub-Drone (Layer 4 continuous mono bed).
    - **Voices 1–2 (Track 2):** Chord Pad Root & 5th (Layer 2 polyphonic pad floor).
    - **Voice 3 (Track 2):** Shared Pad/Lead Voice (borrowed by Lead during active melodic phrases, returning seamlessly to Pad as 9th/triad extension during breathing pauses).
    - **Voice 4 (Track 3):** Shimmer Arpeggio (Layer 3 mono cascade).
    - **Voices 5–7 (Track 2):** Reserved Live Performance Headroom (3 synth voices strictly reserved for real-time play on the 27 keys).
    - **Track 4 (Isolated Drum Engine):** 4 dedicated, isolated drum voices (micro-accents, kick, click, shaker).
    - **Preemption Priority & Live Key Headroom Guarantee:** Voice preemption priority is decoupled from voice ownership: Live Keys (PRIORITY_CRITICAL) > Lead Melodic Voice > Chord Pad Voices > Shimmer Arp > Sub-Drone (LOWEST). Live physical keys are guaranteed never to be preempted by generative voices; if a live key exceeds the 3-voice headroom, it preempts the lowest generative voice with a 10 ms cosine amplitude ramp-down.
    - **Cap Drum Voices at 4 (Layer 1):** Capped at 4 drum voices (reduced from stock 6). Euclidean ambient percussion only fires 2–3 simultaneous micro-accents (kick, woody click, shaker), so capping at 4 eliminates unnecessary CPU spikes during dense polyrhythmic passages. Includes global Percussion Filter Damp and transient softening to gently integrate percussive textures.
    - **Provisional Master FX, Stock SLOOP Reuse & DSP Memory Budgets:** To prevent redundant code, the Step 2 source audit mandates inspecting existing SLOOP effects in `fx.c` (built-in lookahead limiter, distortion, reverb, chorus) to reuse proven DSP. Processing buffers are explicitly budgeted from the 344 KB audio heap (~20.75 KB total): tape wow/flutter delay buffer (50 ms max @ 44.1 kHz, 16-bit mono = ~4.4 KB), lookahead limiter buffer (2 ms lookahead / 88 samples stereo = ~350 bytes), and half-rate diffusion reverb (22.05 kHz, 16-bit mono network = ~16.0 KB). The master bus peak limiter uses lookahead buffering and a smooth release set at -0.5 dBFS; because true-peak inter-sample suppression requires oversampling, this ceiling guarantees sample-peak protection with measured headroom. Audio sub-block timing resolution depends on whether SLOOP's audio callback consumes events at 64-sample block boundaries (~1.45 ms); sample-accurate sub-block offsets are an audit requirement to be verified in Step 2.

    - **Preemption Fade & Amplitude Envelopes:** Voice transitions execute a 10 ms cosine amplitude ramp-down to eliminate preemption clicks during voice reallocation, while chord and modal transitions use a 50–100 ms exponential amplitude release.
    - **Preserve the Reliable 8-Synth Pool:** Rather than risking RAM overflow or ISR starvation by pushing to 9+ synth voices, keep 8 synth voices and resolve harmonic density via dynamic Voice Borrowing (lead voice borrows Voice 3 of the chord pad during phrases, which swells back into a full triad during breathing pauses).
    - **Authoritative Summary:** 8 Synth Voices + 4 Drum Voices = 12 total managed voices across Subsystems 1 & 2.
- **Zero-Crossing Audio Safety & Interpolation:** Smooth voice release windows (50–100 ms exponential fade) are enforced during modal shifts and auto-latch generative hand-offs to avoid zero-crossing clicks. Delay-line read pointer modulations for wow/flutter and thermal portamento utilize linear/hermite interpolation to completely prevent zipper noise.
- **Provisional Master Peak Limiter & Soft-Saturation:** Master bus processing combines tanh soft-saturation with a true sample-peak ceiling placed directly post-saturation to catch inter-sample overshoots, preserving warm tape grit while protecting output levels.
- **RAM & Display Strip-Rendering Architecture:** Static `.data+.bss` RAM has a strict ~98 KB ceiling with only ~5.5 KB of headroom remaining. Allocating a full 240x240 16-bit display framebuffer (~115 KB) exceeds available RAM, and full-screen 240x240 16-bit rendering over SPI takes 23–46 ms, exceeding the 41 ms (24 FPS) rendering budget. The UI avoids full-screen per-pixel watercolor blending, using instead coarse blended color zones and strictly bounded dirty rectangles (orbital arcs, center text, and dials). The 4–8 KB canvas tile is allocated once at boot from the dynamic audio heap, marked non-cacheable/DMA-capable, with zero runtime allocation or freeing to preserve the 5.5 KB BSS RAM. Sub-millisecond event scheduling handles event offsets within 64-sample audio blocks (~1.45 ms at 44.1 kHz) to preserve micro-timing. Furthermore, all generative and modulation math uses 32-bit fixed-point integer arithmetic referencing the `SINE_LUT[1024]` table to prevent floating-point pipeline stalls inside audio interrupts.

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5. Hardware Integration & Real-World Limitations

- **Live Pass-Through & 27-Key Interaction Matrix:** Actively playing the device's 27 silicone keys sends notes directly to the synth engine. White keys 1–16 act as musical performance keys in Slambient live mode with OCT-/OCT+ octave transposition controls, while physical buttons (SEQ, EDIT, FX, GLO, HOME) retain standard UI menu navigation functions.
- **Leaky Pitch-Class Histogram & Mode Inference:** As manual performance notes are played on the 27 silicone keys, a leaky pitch-class histogram analyzes note distribution

in real-time, inferring both the root anchor and scale mode (e.g., Dorian vs. Lydian) from manual play.

- **Auto-Latch Seamless Transition:** When no keys are touched for a brief threshold (e.g., 3 seconds), an internal flag triggers a harmonic hand-off. The engine captures the inferred root and modal scale, seamlessly reassigning them as the new generative anchor for wandering bass, lead, and pads.
- **Bluetooth BLE Hardware Disable:** In custom SLOOP/Felucca firmwares, the Bluetooth BLE radio is intentionally powered off and kept disabled to eliminate RF interference bleeding into the DAC and prevent BLE interrupt latency. The projected 60 KB RAM savings from BLE disablement is unverified pending full linker map analysis in Step 2. All MIDI communication is handled exclusively via class-compliant USB-C MIDI and the 3.5mm TRS hardware MIDI IN jack.
- **Two-Sector Flash Journaling Rotation & Flashing Safety:** Flash persistence uses a two-sector ping-pong rotation scheme (Sector A & Sector B, 4 KB each) on dedicated flash storage. Records are formatted in a compact 24-byte layout consisting of sequence numbers, active world seed, macro parameters, and a rolling CRC32 checksum. The alternate sector is pre-erased during Eno silence pauses or transport STOP so low-battery emergency saves execute strictly as fast 24-byte program writes without erase stalls. Single-bank flashing risks, ROM recovery, and interrupted update behaviors are unconfirmed in reverse-engineering research (AL-255/FM-1-RE); rollback capabilities are experimental until tested on physical hardware, requiring a verified stock V15 image and backups before flashing.
- **Generative "World" Seeds & Determinism Rules:** Entire generative session states are represented as a compact 32-bit xorshift PRNG seed combined with macro parameter settings. The full 32-bit seed is formatted as an 8-character hex code (e.g., `#A3F2-B8D4`). An all-zero seed is strictly forbidden and automatically remapped to `0x534C414D`. A 4-character truncated shorthand (e.g., `#D7A4`) is used purely for display-only compact UI views and is never used for preset recall.
- **Physical Hardware Knobs & External MIDI Sync:** Dedicated physical knobs provide direct tactile control (Knob 1: Mutation Density, Knob 2: Drone Weight, Knob 3: Shimmer Speed, Knob 4: Spatial Diffusion/Reverb). The SELECT macro knob ("Weather") smoothly interpolates overall density, BPM (40–75 BPM), drone overtone weight, and thermal portamento drift. When an external MIDI clock signal is detected, the "Weather" macro's internal BPM modulation automatically decouples—keeping the external clock authoritative while continuing to modulate density, drone weight, and portamento.
- **Layer 1 Rhythmic Evolution:** To ensure the 7-step Euclidean drum pattern continuously evolves without feeling repetitive, Layer 1 incorporates per-bar probability gating (chance), dynamic fill muting, and subtle pattern rotations every 2 or 4 bars.
- **Power-of-Two Drop-on-Full FIFO Decoupling:** The lossy UI display FIFO buffer is sized to 32 entries (power-of-two) and drops incoming visual events on full, ensuring the display thread never stalls the audio ISR. Critical audio event paths (note-offs, voice releases, panic commands) bypass the lossy UI FIFO entirely and execute directly via guaranteed atomic event handlers.

- **Audio I/O Specification & USB Audio Direction:** The 3.5mm analog jack operates as stereo audio output only (with automatic internal speaker mute). Stock SLOOP USB Audio operates strictly as a USB Audio Device Class output to host (allowing a host computer to record the FM-1 master output). External audio input into the FM-1 from a host is unverified and flagged as an unconfirmed dependency; external-input DSP processing will not be planned until verified in source code.

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6. Modular C Firmware Structure (`slambient-fm1`)

The codebase is cleanly modularized into three decoupled subsystems, aligning with the implementation plan across `slambient.h / slambient.c` (Subsystem 1: Generative Engine), `audio.c / seq.c / voice.c` (Subsystem 2: Synth & Sequencer Engine), and `slambient_fx.c / ui_studio.c / presets.c` (Subsystem 3: Optional Audio DSP, UI & Flash Persistence):

- **`slambient.h / slambient.c (Subsystem 1):`** Core Generative Engine managing 6-prime clock dividers, scale math, stateful random-walk stepping, modal drift rules, micro-tunings, thermal portamento, mood presets, and lock-free FIFO ring buffer queuing to Subsystem 3.
- **`audio.c / seq.c / voice.c (Subsystem 2):`** Existing Synth & Sequencer Engine handling track routing, note event execution, 8-synth/4-drum voice allocation table, amplitude envelopes, and live 27-key silicone keyboard pass-through.
- **`slambient_fx.c / ui_studio.c (Subsystem 3):`** Optional Audio DSP and Display Engine managing provisional master bus tape wow/flutter, soft-saturation, lookahead peak limiting, and non-blocking 24 FPS strip-rendered display updates.
- **`presets.c / .h`**: Manages onboard storage, recall of custom ambient user slots, instant mood profiles, and Auto-State Snapshots with pre-erased flash journal sector management.

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7. Slambient Hardware UI, Synesthetic Engine & TFT Visualizer

- **Boot Screen:** Custom minimalist "Slambient" startup animation with a breathing analog wave pattern.
- **Synesthetic Color Mapping Engine:** Pitch classes, scale modes, and chord intervals dynamically correspond to specific color coordinates (e.g., deep earthy roots mapping to warm amber/ochre, ethereal thirds/fifths shifting into twilight indigos, and sparkling arpeggios adding vibrant moss greens or celestial cyans).
- **Harmonic Color Blending:** As notes overlap across the 4 layers, the TFT display real-time blends these chromatic values together across the UI elements, creating an evolving, living watercolor effect that mirrors the audio.

- **TFT Center Display:** Bold, clear text showing the active root note and chord voicing (e.g., `D : WARM_TRIAD`) rendered in the primary synesthetic color of the chord.
- **Harmonic Orbit & Layer Arcs:** A morphing organic center polygon accompanied by four concentric vector arcs representing the 4 prime-number layers as they slowly drift and phase-lock over time.
- **Battery Conservation, Auto-Dim & Diagnostic Instruments:** Includes a 60-second display auto-dimmer that drops backlight brightness to 30% during generative idle, extending battery runtime to 10–12 hours. A development overlay provides real-time diagnostic instruments including an audio buffer underrun counter and a CPU load percentage meter.
- **TFT Display Architecture & Refresh Rate Benchmarking:** Screen updates utilize non-blocking DMA framebuffer transfers and dirty-rectangle rendering for the harmonic orbit polygon, concentric arcs, and color blending. Real SPI DMA frame durations will be benchmarked on physical hardware under audio load during Step 7 to establish actual achievable frame rates rather than assuming 24 FPS is guaranteed.

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8. Hardware Verification & Pre-Flight Checklist

- **Firmware Baseline & Pinned Toolchain:** The firmware baseline is strictly pinned to SLOOP v2.4/v2.5 (isod89/sloop-fm1 repository), JieLi toolchain version `jieli-linux-toolchains-20260730.1` via Docker Desktop, and SDK `fw-AC79_AIoT_SDK`.
- **Target Binary Package:** Compiles to `build/felucca.fwsc` with package ID `FM-1_900`.
- **Flashing Procedure & Safety:** Flashed via USB-C using WebMIDI in Chrome/Edge at fm1.designburgapps.com. Because single-bank flashing risks remain unconfirmed, rollback functionality is experimental and requires maintaining verified stock V15 binaries prior to flashing.
- **Desktop Simulator First & Golden-Seed Regression Tests (Step Zero):** Prior to deploying to physical hardware, compile `generator.c` and `sequencer.c` as a standalone macOS command-line executable outputting MIDI directly to a DAW or via RtMidi. Simulator determinism relies on an abstracted virtual clock with recorded timeline inputs, eliminating unseeded rand() calls or real-time jitter. Automated golden-seed CLI tests verify that identical world seeds produce byte-identical event streams across multi-hour runs before flashing firmware.

9. Prioritized 8-Step Engineering Roadmap & Implementation Plan

This section outlines the authoritative 8-step build order for engineering, auditing, integrating, and verifying the Slambient firmware on physical hardware.

- ### Step 1: Prove Firmware Recoverability & Pin Baseline
  - Pin baseline to SLOOP v2.4/v2.5, JieLi toolchain (jieli-linux-toolchains-20260730.1), and fw-AC79_AIoT_SDK. Compile stock binary, verify clean build, measure BSS/flash

memory, and verify USB WebMIDI flashing pipeline and firmware recoverability with stock V15 backups.

Step 2: Source Code & Hook Audit

- Inspect `audio.c`, `fx.c`, `seq.c`, and `voice.c`. Audit post-mix audio bus tap points, sample-accurate sub-block event offsets, cross-track voice allocator vs. per-track fallback, USB audio direction, memory map, and stock FX reuse in fx.c.

Step 3: Desktop Simulator Verification (macOS Native CLI)

- Compile Subsystem 1 (generator.c, sequencer.c) natively on macOS with RtMidi output. Run automated statistical regression suite across 1,000,000 steps to verify 10% tension target, zero voice leaks across the 8-voice table, voice allocation invariants, and deterministic golden-seed output.

Step 4: Single-Layer Hardware Integration

- Integrate Subsystem 1 into embedded stock codebase driving Layer 2 (Pad/Lead) only. Verify event trigger reliability, 27-key silicone keyboard live pass-through on Track 2, and smooth exponential amplitude releases on modal transitions.

Step 5: Multi-Layer Expansion & 8-Voice Accounting Stress Test

- Enable all 4 generative layers (Drone, Chords/Lead, Arp, Drums). Execute rigorous stress testing of the authoritative voice allocation table (Voices 0–4 generative, Voices 5–7 live keys, Track 4 drums). Verify Voice 3 dynamic borrowing, Stuck-Note Watchdog timeouts, and zero voice starvation during heavy live play.

Step 6: Master DSP & Audio Tap Insertion

- Insert Subsystem 3 master audio processing into audio callback (evaluating stock FX reuse from fx.c). Measure latency, CPU load, and audio buffer underruns on hardware within the 20.75 KB DSP buffer budget.

Step 7: 24 FPS Strip-Rendered UI & Flash Persistence

- Connect Subsystem 1 event FIFO to Subsystem 3 display thread. Implement non-blocking frame renderer, benchmark SPI DMA frame duration, and implement two-sector ping-pong flash persistence with pre-erased sector rotation.

Step 8: Physical Hardware Verification & Long Soak Testing

- Perform continuous 24-hour burn-in soak test on physical hardware. Monitor thermal stability, audio buffer underrun counters, battery power drop-off behavior, pre-erased flash journaling integrity, and USB MIDI log parity against simulator golden seed outputs.